HEFTNet: modelling the di-Higgs m_{hh} spectrum and its uncertainty

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Executive summary and roadmap, October 31, 2026

Goal. A fast surrogate for POWHEG: the di-Higgs cross section as a function of m_{hh} (255–1455 GeV) and of five HEFT coefficients, with a calibrated uncertainty, written as $d\sigma/dm_{hh} = \sum_{i=1}^{23} C_i(\theta)\, a_i(m_{hh})$ with known polynomials C_i and 23 functions a_i of m_{hh} alone. Data: 3,989 POWHEG spectra for the fit; 1,000 independent test spectra for validation.

Principal findings

1. **The primary model is the smooth 23×80 model**: each $a_i(m_{hh})$ is a smoothed spline times the phase-space factor, fitted to the bin *integrals* by Poisson maximum likelihood, with smoothness set by the evidence. It is linear in its 920 parameters, so its posterior is exactly Gaussian; no training, a minute on a laptop. It is a genuine density in m_{hh}: fitted on 30-GeV bins it predicts the 15-GeV split to within the noise above 360 GeV.
2. **It is as accurate as any model of this form can be.** A free fit of 23 numbers per bin, the floor for the form $\sum_i C_i a_i$, does no better (likelihood-ratio statistic $-2\ln\lambda$ per bin 1.56/1.10 below/above 555 GeV for both); neither does the network ensemble (1.56/1.12).
3. **The data are exact Poisson counts**, $q_s\times$ integers (40,000 events per run), and weighting by the *observed* counts biases fits low; weighting by the model's Poisson variance (iterated) removes the bias.
4. **The residual model error comes from cancellations.** Spectra whose cross section is a small difference of large terms deviate smoothly from the 23-term form, in POWHEG itself. A discrepancy $\varepsilon(m_{hh})\sum_i |C_i a_i|$, with $\varepsilon = 0.08\%$ at threshold to 3.7% at 1.45 TeV, fitted on rows not used in the fit, gives pull s.d. 1.004 (1.011/1.001 below/above 555 GeV), 95.4% within ± 2, and calibration at every degree of cancellation. The fit uncertainty itself is 2.9% of the noise.
5. **Cross-checks agree.** A 20-network ensemble (L1 + Huber-pull fine-tune) gives the same accuracy and, with the same discrepancy, the same calibration; its spread is 80% optimisation. A Bayesian network by HMC undersampled (0.5% of the noise); a mixture of Laplace approximations gives 2.4% $\oplus$ 5.6%, consistent with the smooth model.

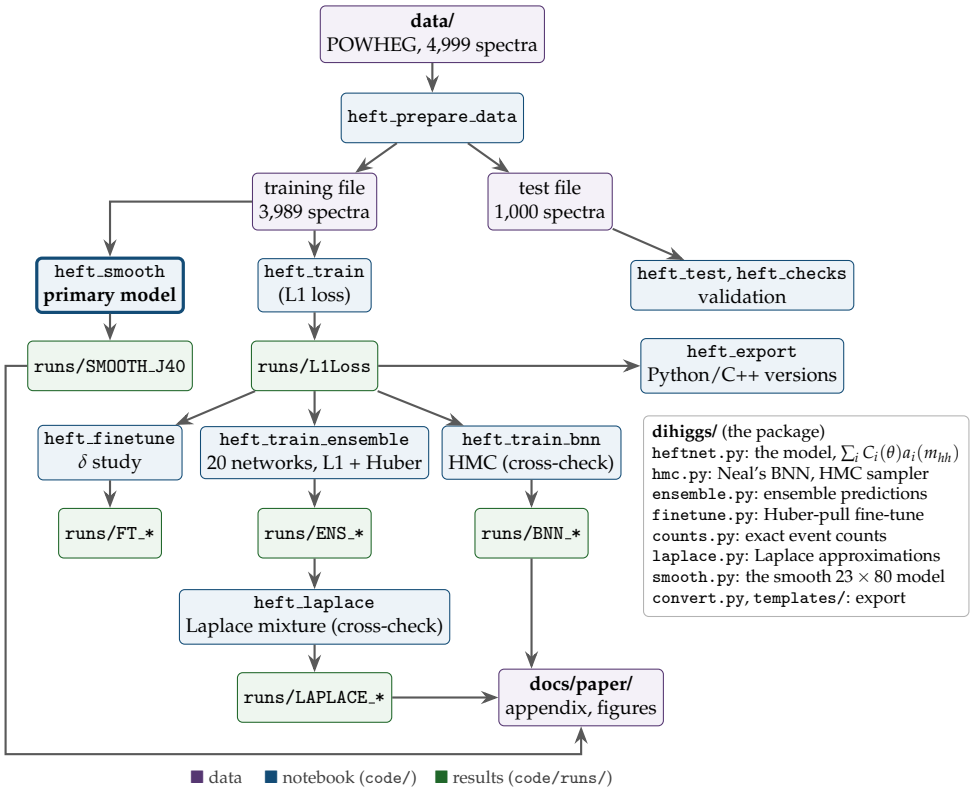
Investigations, in order.

- *BNN with HMC* (`hmc.py`); diagnosis in `docs/bnn/bnn_plan2.pdf`.
- *Ensembles* (`ensemble.py`): seeds vs bootstrap, reference ensembles (`docs/ensemble`).
- *Huber-pull fine-tune* (`finetune.py`), `docs/huber`.
- *Exact counts* (`counts.py`), `docs/counts`.
- *Mixture of Laplace approximations* (`laplace.py`), `docs/bnn/laplace_explained.pdf`.
- *Origin of the model error*: free-fit benchmark, cancellation-aware discrepancy (`docs/modelerror`).
- *Smooth* 23×80 *model* (`smooth.py`), `docs/smooth`.

Result for the paper. The smooth model with the cancellation-aware discrepancy; the network ensemble and the Bayesian network as cross-checks. Draft appendix and figures: `docs/paper`.

Open items. A discrepancy correlated in m_{hh} (Gaussian process); the origin of the cancellation-dependent deviations and of the rise of ε at high mass (questions for the generator side); finer knots near threshold; use only within 250–1455 GeV.

The `dihiggs` repository at a glance



Roadmap of the documents (all in `docs/`; read top to bottom, or jump to the topic you need)

Start here	
`paper/executive_summary.pdf`	this summary
Physics and the model	
`theory.pdf`	LHC Higgs WG note on EFT descriptions of hh production (arXiv:2304.01968): the C_i
`dihiggs_model.pdf`	the `HEFTNet` model: exact θ dependence, a one-dimensional learning problem
`spectra.pdf`	one-page overview of the POWHEG spectra
Data, pipeline and checks	
`report/dihiggs_report.pdf`	the pipeline from POWHEG files to trained and tested model
`checks_plan.pdf`	three checks of the data and test pipeline
`counts/counts_note.pdf`	the spectra are scaled Poisson counts: exact noise
Training	
`huber/huber_plan.pdf`	the Huber-pull fine-tune (the accuracy gain near threshold)
`curriculum_plan.pdf, _v2.pdf`	side study: curriculum learning
The primary model	
`smooth/smooth_plan.pdf`	design of the smooth 23×80 model
`smooth/smooth_results.pdf`	its results: accuracy, calibration, within-bin and edge tests
`modelerror/model_error_note.pdf`	where the model error comes from (free-fit benchmark)
`modelerror/new_error_model.pdf`	the cancellation-aware discrepancy
Uncertainty (network studies)	
`ensemble/ensemble_plan.pdf`	seeds and bootstrap ensembles; the reference-ensemble question
`bnn/bnn_plan.pdf`	design of the Bayesian network and the HMC module
`bnn/bnn_run1.pdf–run3.pdf`	the three HMC runs (incl. a primer on mixing)
`bnn/bnn_plan2.pdf`	why HMC gave too-small widths; the Laplace plan
`bnn/laplace_explained.pdf`	the mixture of Laplace approximations, explained, with results
Paper	
`paper/appendix_draft.pdf`	draft JHEP appendix: the model and its uncertainty
`paper/fig_*.pdf`	paper figures: 9 spectra (`fig_spectra_panel`); the 23 $a_i(m_{hh})$ (`fig_coefficients`)
`paper/all_test_spectra.pdf`	all 1,000 test spectra against the smooth model (51 pages)
Practical	
`remote_gpu.pdf`	running the notebooks on the RTX 5090 (`tmux`, `nohup`)